Florian Hébert

Data Scientist

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Building robust and reliable data pipelines and computer vision algorithms to solve high-stakes problems, from proof-of-concept to deployment into production. Proven ability to work effectively with scientific and technical stakeholders and software development teams to develop and deploy impactful solutions.

Work experience

▶ **Bioinformatics Engineer** 06/2022 –
Stilla Technologies / Bio-Rad LSG Paris Villejuif, France

- Eliminated the main cause of chip production failure by designing a new image analysis stack integrated in the commercial software; enabled to safely launch a new chip production line.
- Developed a region-of-interest localization method that replaced the existing algorithm in the commercial software, resolving all previously known failure cases and ensuring 100% reliability in internal validation.
- Developed and deployed automated defect detection tools used by 10+ engineers, improving root cause analysis for 5 defect types.
- Built an interactive image annotation web application used by 20+ users, structuring scattered and heterogeneous data.
- Pioneered the calculation of advanced biological metrics, resulting in new features integrated in the commercial software and used namely in oncology and gene therapy research.
- Implemented a pipeline to train neural networks for image classification. Used to train CNNs integrated in the commercial software, reaching 98% in both true positive rate and precision.
- Developed a new vignetting correction method, rescuing 10+ instruments from QC failure.

▶ **R&D Data Scientist** 11/2019 – 06/2022
Microbs Rennes, France

- Pioneered the development of monitoring and analysis tools cutting the analysis time by 90% and enabling the R&D team to release a total viable count test.
- Trained and evaluated bacteria recognition algorithms integrated in 10+ released instruments, reaching 95% true positive rate and a near-zero false positive rate.

Education

▶ PhD Thesis (Applied Statistics), Agrocampus Ouest, Rennes, France, 2019
▶ Master's Degree, Applied Statistics / Data Science, Rennes 2 University, France, 2016

Technical skills

▶ **Computer vision:** OpenCV, TensorFlow, CNN (image classification, image segmentation)
▶ **Programming languages:** Python (NumPy, Pandas, Scikit-Learn, TensorFlow), SQL, R
▶ **Production and deployment:** Git, Docker, Docker Compose, automated testing

Languages

▶ **French:** Native ▶ **English:** Proficient